

Summary

In this project, we studied partitions of integers subject to constraints by

- ▶ constructing bijections,
- ▶ writing computer programs,
- ▶ using combinatorial tools called *generating functions*.

Integer partitions

Let’s list the different ways we can sum positive integers to 6:

6,

4 + 1 + 1,

3 + 1 + 1 + 1,

2 + 1 + 1 + 1 + 1,

5 + 1,

3 + 3,

2 + 2 + 2,

1 + 1 + 1 + 1 + 1

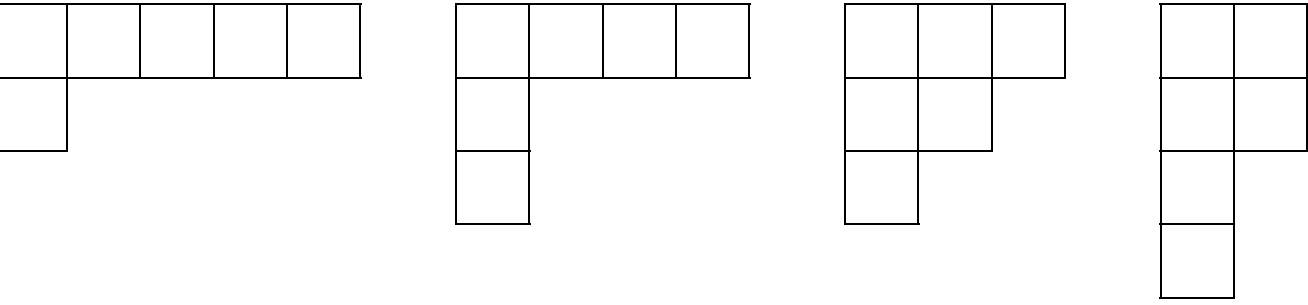
4 + 2,

3 + 2 + 1,

2 + 2 + 1 + 1,

1 + 1 + 1 + 1 + 1 + 1

We’re ignoring the order, and we allow the “empty” partition 6. A visual aid is a *Ferrers board*:



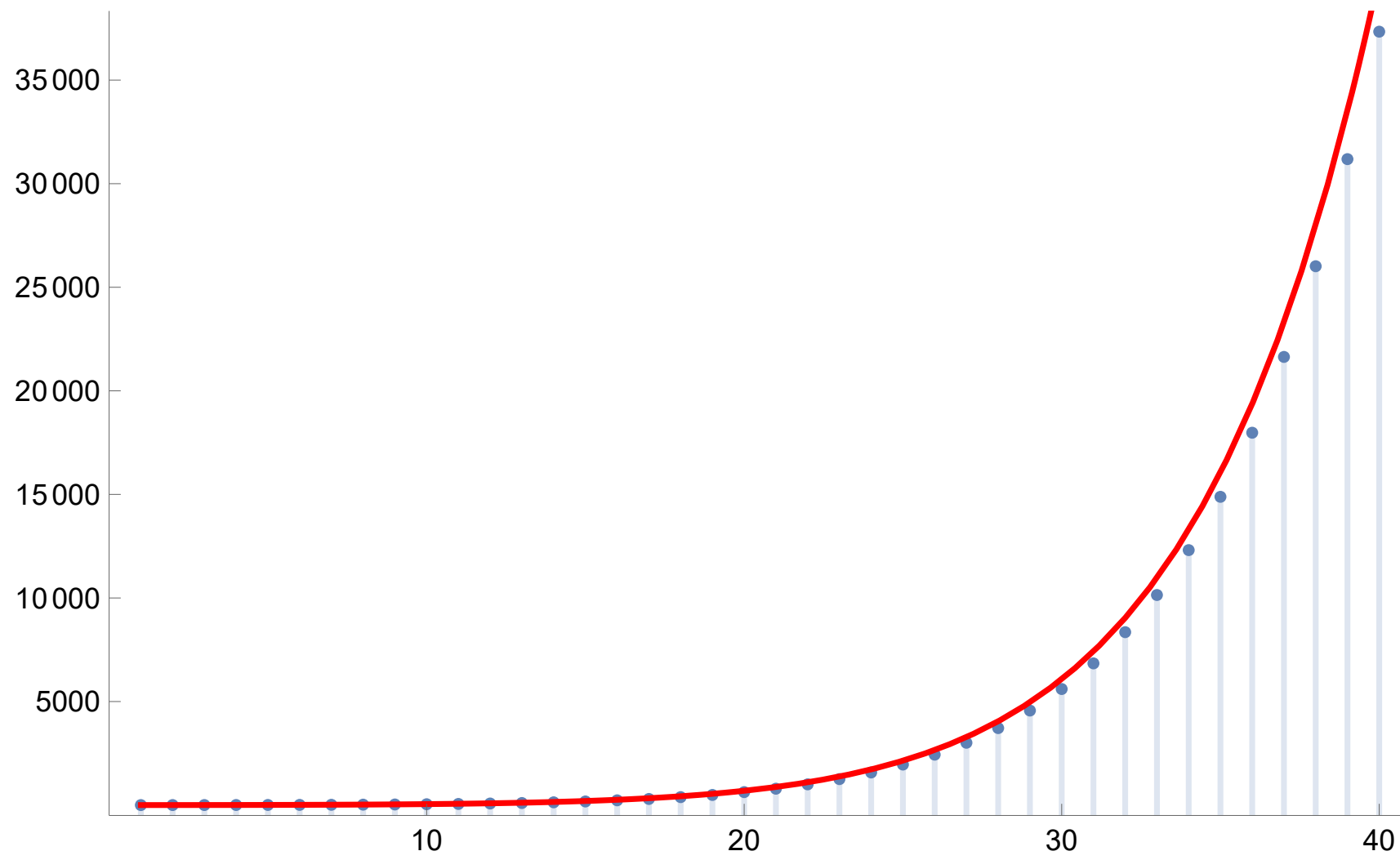
Which partition of 6 listed above does each board correspond to? It’s hard to compute all the partitions of a large integer by hand. To help, we wrote some computer programs in Python that generate a list of the partitions of n for any input integer n .

Counting the partitions

By counting the ways of partitioning any integer n , we define the *partition function*

$$p(n) = \# \text{ of partitions of } n.$$

For example, $p(6) = 11$ by counting the items in the list above. $p(n)$ grows rapidly as $n \rightarrow \infty$; for example, $p(40) = 37,338$. To compute $p(n)$ for large n , it’s not necessary to compute all the partitions of n . Instead, we wrote another program using a dynamically programmed recursive scheme. This relies on a consequence of Euler’s pentagonal number theorem (later).



$p(n)$ (the blue dots) plotted for $n \leq 40$

Theorem (*Hardy and Ramanujan, 1918*):

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right) \text{ as } n \rightarrow \infty.$$

In the picture above, the right-hand side of Hardy and Ramanujan’s estimate appears in red. You can see that it follows $p(n)$ closely, even for small values of n . Hardy and Ramanujan introduced the now-famous *circle method* to prove this result, and we spent a week learning about this technique.

Partition identities

Rather than listing all partitions of a given n , we can impose a condition on the parts. For example, if we select the partitions of 6 that have odd parts, we obtain

5 + 1,

3 + 3,

3 + 1 + 1 + 1,

1 + 1 + 1 + 1 + 1

This defines a *restricted partition function*

$$p(n \mid \text{odd parts}) = \# \text{ of partitions of } n \text{ with odd parts,}$$

and we now know that $p(6 \mid \text{odd parts}) = 4$. Alternatively, the partitions of 6 whose parts are distinct (no repetitions allowed) are

6,

5 + 1,

4 + 2,

3 + 2 + 1,

so $p(6 \mid \text{distinct parts}) = 4$. Notice that this is the same as $p(6 \mid \text{odd parts})$! In 1748, Euler proved that

$$p(n \mid \text{odd parts}) = p(n \mid \text{distinct parts})$$

for *all* n . We learned two proofs of this result; one is bijective, and the other uses generating functions. For two different conditions A, B on parts of partitions, an equation of the form

$$p(n \mid A) = p(n \mid B),$$

for all n , is called a *partition identity*. Studying these sorts of equations was the main object of our project.

Bijective proofs of partition identities

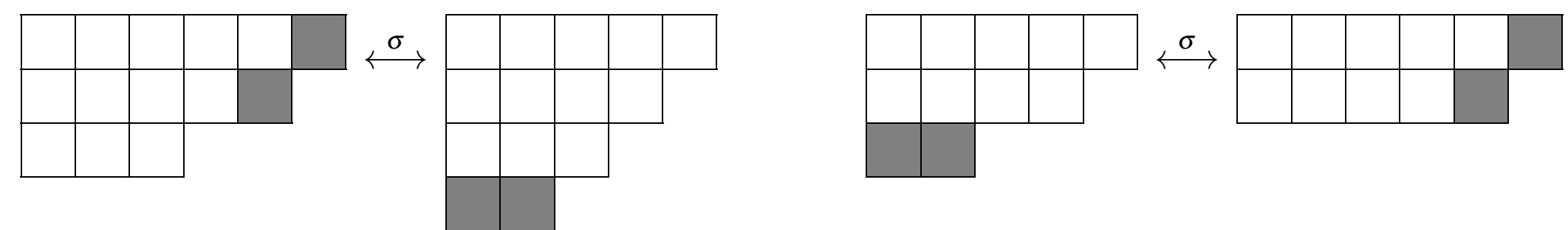
To prove certain partition identities, it suffices to define bijections between sets of Ferrers boards modeling partitions described in the identity. These bijections also tend to provide more insight into the nature of certain identities. For example, Eulers pentagonal number theorem states that

$$p(n \mid \text{even \# of distinct parts}) = p(n \mid \text{odd \# of distinct parts}) + e(n),$$

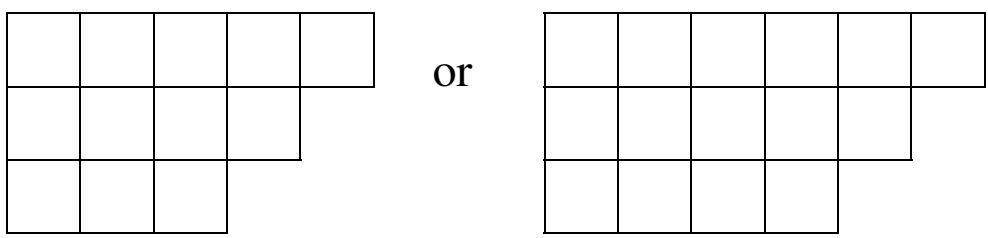
where

$$e(n) = \begin{cases} (-1)^j & \text{if } n = \frac{1}{2}j(3j \pm 1) \text{ for } j \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases}$$

In 1881, Fabian Franklin found a bijective proof of this identity. His bijective involution σ relates partitions with an even/odd number of distinct parts to other distinct partitions which are odd-/even (of opposite parity). σ is defined by the following rule: if the board’s ‘rightmost diagonal’ $<$ the board’s smallest row, it becomes the new smallest row. Otherwise, distribute the boxes of the bottom row into the other rows.



This involution provides the one-to-one correspondence between the desired sets except in the case when the rightmost diagonal intersects the bottom row and the bottom row is one larger than this diagonal. This only can only occur for a partition of the pentagonal numbers (as seen below), providing the exception defined in the error function $e(n)$:



Generating functions

The generating function $f(x)$ of a sequence $a_0, a_1, a_2, \dots$ is

$$f(x) = \sum_{i=0}^{\infty} a_i x^i = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

Here x is treated as a placeholder to keep track of the coefficients. The sequences we are interested in are sequences of values of partition functions subject to a condition A :

$$p(0 \mid A), p(1 \mid A), p(2 \mid A), \dots$$

with the convention $p(0) = 1$. The corresponding generating function traditionally uses the variable q instead of x :

$$\sum_{n \geq 0} p(n \mid A) q^n = p(0 \mid A) + p(1 \mid A) q + p(2 \mid A) q^2 + \dots$$

By manipulating series of this form, we can prove identities of the form

$$\sum_{n \geq 0} p(n \mid A) q^n = \sum_{n \geq 0} p(n \mid B) q^n,$$

which implies that $p(n \mid A) = p(n \mid B)$ for all n .

A proof using generating functions

We proved two equations of Euler, which give generating functions for two restricted partition functions:

$$\sum_{n=0}^{\infty} p(n \mid \text{distinct parts}) q^n = \prod_{n=1}^{\infty} (1 + q^n)$$
$$\sum_{n=0}^{\infty} p(n \mid \text{odd parts}) q^n = \prod_{n \text{ odd}} \frac{1}{(1 - q^n)}$$

We also proved that these two infinite products are in fact equal, which shows that

$$p(n \mid \text{distinct parts}) = p(n \mid \text{odd parts})$$

for all n . The first several terms of these infinite products are

$$1 + q + q^2 + 2q^3 + 2q^4 + 3q^5 + 4q^6 + 5q^7 + 6q^8 + 8q^9 + \dots$$

The coefficient of q^n gives the number of partitions of n into odd (or distinct) parts. Notice that the coefficient of q^6 is 4, which we calculated previously.

The Rogers-Ramanujan identities

Rogers (independently Ramanujan) proved that

$$p(n \mid \text{all parts differ by at least 2}) = p(n \mid \text{all parts} = 1 \text{ or } 4 \pmod{5})$$

For example, the partitions of 6 whose parts differ by at least 2 are

6,

5 + 1,

4 + 2,

and the partitions whose parts are $= 1$ or $4 \pmod{5}$ are

6,

4 + 1 + 1,

1 + 1 + 1 + 1 + 1 + 1.

In both cases, there are 3 such partitions. There is a companion identity which additionally restricts the smallest part in the left condition to be ≥ 2 , and the residues of $1, 4 \pmod{5}$ are replaced by $2, 3 \pmod{5}$ on the right.

Generating the Rogers-Ramanujan identities

In the language of generating functions, the first Rogers-Ramanujan identity reads

$$\sum_{n=0}^{\infty} \frac{q^{n^2}}{(q; q)_n} = \prod_{m=0}^{\infty} \frac{1}{(1 - q^{5m+1})(1 - q^{5m+4})} = 1 + q + q^2 + q^3 + 2q^4 + \dots$$

where $(a; q)_n = (1 - a)(1 - aq) \dots (1 - aq^{n-1})$ is the *q-Pochhammer symbol*. There is a similar formula for the second identity. We spent a few weeks translating the Rogers-Ramanujan identities into generating functions, and then establishing equality. We related these identities by first establishing some recurrence relations for finite q -series, and then took limits. Our proof also crucially relied on the *Jacobi triple product identity*:

$$\sum_{n=-\infty}^{\infty} z^n q^{n^2} = (q^2; q^2)_{\infty} (-zq; q^2)_{\infty} (-z^{-1}q; q^2)_{\infty}$$

The Gordon-Göllnitz identities

In terms of generating functions, these two identities are

$$\sum_{n=0}^{\infty} \frac{q^{n^2} (-q; q^2)_n}{(q^2; q^2)_n} = \prod_{m=0}^{\infty} \frac{1}{(1 - q^{8m+1})(1 - q^{8m+4})(1 - q^{8m+7})}$$
$$\sum_{n=0}^{\infty} \frac{q^{n^2+2n} (-q; q^2)_n}{(q^2; q^2)_n} = \prod_{m=0}^{\infty} \frac{1}{(1 - q^{8m+3})(1 - q^{8m+4})(1 - q^{8m+5})}$$

The first identity states that the number of partitions whose parts differ by at least two, and for which no consecutive even parts appear, is equal to the number of partitions whose parts are congruent to 1, 4, or 7 (mod 8). The second identity adds to the difference condition that the smallest part is ≥ 3 , and the residues (mod 8) in the second condition are replaced by 3, 4, and 5. We gave two proofs of this result, based on a paper of Alladi. The first uses an ingenious splitting of a famous identity of Gauss into even and odd parts, a lot of algebraic manipulation, and an application of Jacobi’s triple product identity. The second follows from a similar splitting of the generating functions in Euler’s pentagonal number theorem, and an application of a *quintuple* product identity.

Conclusion

There is a lot more of partitions to explore; undoubtedly there are many partition identities yet to be discovered. Asymptotics of the restricted partition function $p(n \mid A)$ for various conditions A also seems to be an interesting (and difficult) problem. Integer partitions are also relevant to other fields; they show up in representation theory, singularity theory, atomic physics, and many others.

References

Andrews, George and Eriksson, Kimmo, *Integer Partitions*, Cambridge University Press, 2004, New York NY.

Alladi, Krishnaswami *Some new observations on the Göllnitz-Gordon and Rogers-Ramanujan identities*. Trans. Amer. Math. Soc. **347**(3) (1995), 897–914.